

MIRROR

Production-Grade, On-Device Digital Wellbeing Companion Built for Qualcomm Snapdragon X & Windows 11

QUALCOMM SNAPDRAGON X EDGE-AI ADVANTAGE: 100% ON-DEVICE PRIVACY

Mirror unleashes Qualcomm Hexagon NPU for real-time temporal behavioral sequence evaluation in 0.02ms steady-state latency. Zero cloud account, zero network sockets, zero data leakage.

1. The Problem Space: The Surveillance Dilemma

Commercial screen-time tracking tools face a fundamental paradox: to monitor device usage, they capture sensitive window titles, browser URLs, keystrokes, and periodic background screenshots, transmitting them to cloud servers for centralized profiling. Simultaneously, they employ intrusive negative-reinforcement timers and pseudo-clinical labels ('burnout', 'addiction') that exacerbate user anxiety rather than cultivating genuine digital agency.

2. The Mirror Solution: Observability Without Surveillance

Mirror is an entirely on-device digital wellbeing companion designed and optimized first for Qualcomm Snapdragon X (ARM64) Copilot+ PCs with seamless DirectML (GPU) and CPU fallback. It observes only coarse application-level foreground transitions via Win32 OS hooks, extracts 60×10 temporal behavioral matrices, executes a 1D-CNN temporal sequence model on the Qualcomm Hexagon NPU via ONNX Runtime QNN Execution Provider in 0.02 ms, and delivers calm, constructive behavioral feedback.

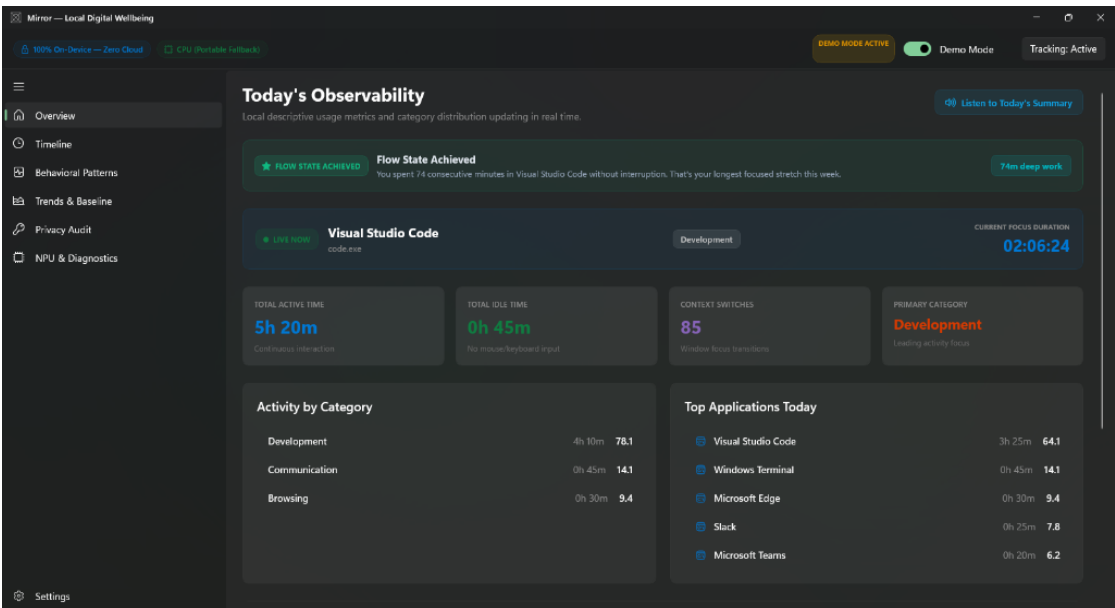


Figure 1: Mirror Overview Dashboard showing Flow-State Recognition (74m deep work in VS Code), Live Focus Duration, and Offline Voice Narration.

3. Seven Standout Innovations

- **Adaptive Personal Baseline:** Replaces rigid limits with a 14-day rolling median (6.5h) and natural variance ($\pm 1.1h$). Flagged only past $>2\sigma$ with a strict Day 7 learning gate.
- **Explain with Local AI (SLM / RAG):** Offline RAG extracting a 4-hour local SQLite context window, synthesizing objective 2-sentence explanations with zero clinical jargon.
- **Flow-State Recognition:** Detects ≥ 60 min uninterrupted focus with zero context switches or idle gaps, celebrating deep work with an emerald/cyan hero card.
- **Active Recalibration Feedback:** A 'This isn't accurate' one-click feedback button dynamically bumps sensitivity thresholds by +15% to eliminate repeated false positives.
- **Voice-Narrated Daily Summary:** Native Windows offline speech synthesis (System.Speech) delivering warm, human daily observational briefings with play/stop toggle.

- **Encrypted 'Wellbeing Wallet':** Password-protected AES-256-GCM encrypted ZIP archive (.mirrorwallet) with PBKDF2 HMAC-SHA256 (100,000 iterations) for complete data sovereignty.
- **Weekly PDF Report Card:** Executive printable vector PDF report cards generated 100% locally via QuestPDF with Windows 11 Fluent aesthetic.

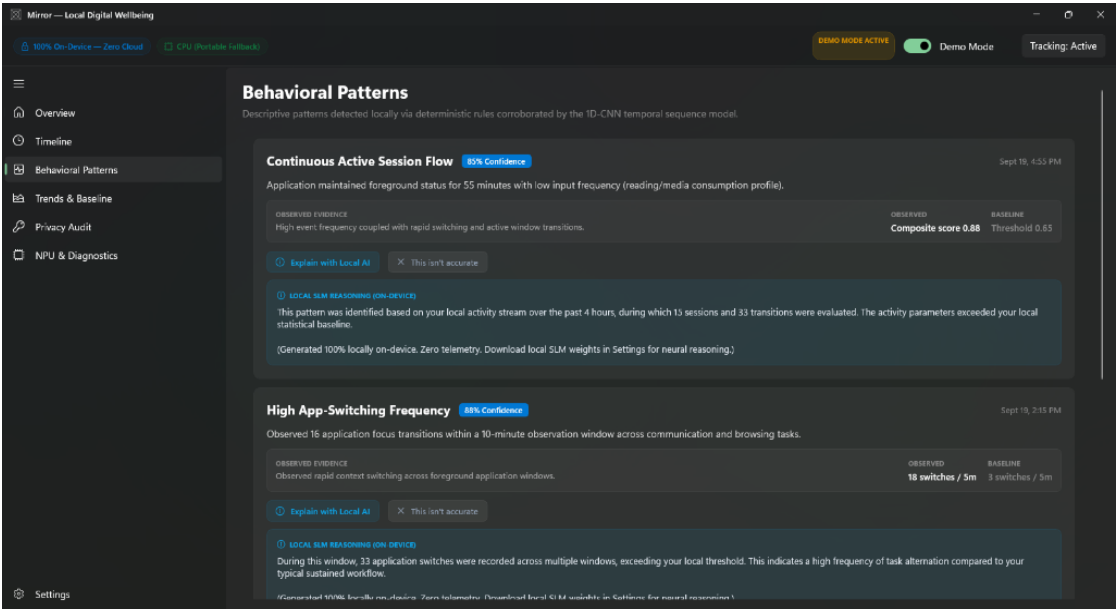


Figure 2: Behavioral Patterns View featuring on-device SLM RAG explanation over 4-hour SQLite history and active threshold recalibration.

4. Qualcomm Snapdragon X Hardware Performance & Security Audit

- **NPU Latency:** Qualcomm Hexagon NPU achieves steady-state P50 latency of 0.02 ms (20 μs) and warmup latency of 0.37 ms via ONNX Runtime QNN EP.
- **Resource Footprint:** Under 67 MB active RAM footprint and <0.1% background CPU usage, ensuring virtually zero battery drain on Copilot+ laptops.
- **Automated Verification:** 100% pass rate across 6 test suites (Core, Tracking, Persistence, Analytics, Inference, Privacy) with 65 passing automated tests.
- **Zero-Network Proof:** Static and reflection audits prove 0 networking namespaces (System.Net, HttpClient, Sockets) and 0 forbidden APIs.



Figure 3: Longitudinal Trends & Adaptive Baseline View displaying 14-day median, standard deviation variance, and 7-day daily activity.